(a) How the reliability check works

1. Error per patient: the share of that patient's comparable pairs the model orders wrongly (censoring-aware).



2. Label high error at a threshold (median, 70th, 80th or 90th percentile).



3. MED3pa learns rules on discovery patients from clinical or pathomic features; each rule defines a profile.

rule on a feature

60% discovery

4. On held-out patients: is error predictable, does the model's C-index differ inside each profile, and do per-patient bounds hold by subgroup at three time horizons?

40% held out:

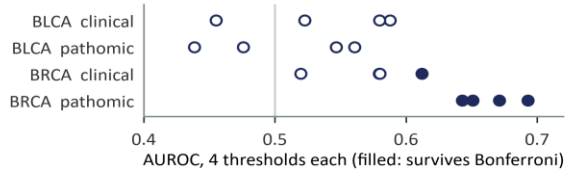
- AUROC vs 0.5
- C-index in group
- 90% bound coverage

TWO DIFFERENT CLAIMS

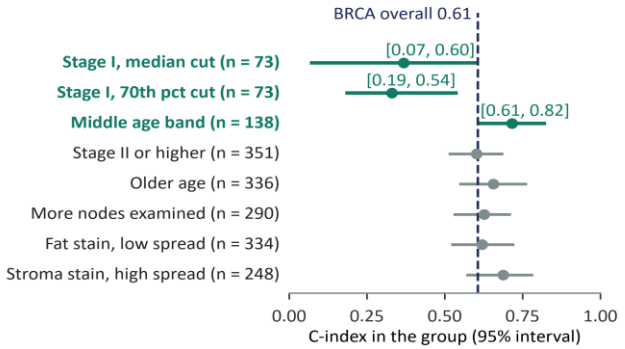
"Error is predictable here" is not "the model performs differently here"; the audit tests each separately.

(b) What the audit found

Is error predictable? In BRCA, yes



Different C-index in a group? 3 of 159 (8 BRCA shown)



Do per-patient 90% bounds hold by stage? (BRCA)

